

Algebra II with Trigonometry

Chapter 11.2

Olympiad Solutions (GPT-5.4)

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[4666322_alg-2trig-h-chp-11-2-note-docx.pdf](#)

Solutions

1. An ellipse is centered at the origin with foci on the x -axis. Its eccentricity is $\frac{3}{5}$, and it passes through $(4, 3)$. Find its equation.

Since the foci are on the x -axis, the ellipse has the form

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad a > b > 0,$$

with

$$e = \frac{c}{a} = \frac{3}{5}.$$

Hence

$$c = \frac{3}{5}a.$$

Using $c^2 = a^2 - b^2$,

$$a^2 - b^2 = \frac{9}{25}a^2 \quad \Rightarrow \quad b^2 = a^2 \left(1 - \frac{9}{25}\right) = \frac{16}{25}a^2.$$

Thus

$$b = \frac{4}{5}a.$$

Now $(4, 3)$ lies on the ellipse, so

$$\frac{16}{a^2} + \frac{9}{b^2} = 1.$$

Substitute $b^2 = \frac{16}{25}a^2$:

$$\frac{16}{a^2} + \frac{9}{(16/25)a^2} = 1$$

$$\frac{16}{a^2} + \frac{225}{16a^2} = 1.$$

Multiply by $16a^2$:

$$256 + 225 = 16a^2$$

$$481 = 16a^2$$

$$a^2 = \frac{481}{16}.$$

Then

$$b^2 = \frac{16}{25} \cdot \frac{481}{16} = \frac{481}{25}.$$

Therefore the equation is

$$\boxed{\frac{x^2}{481/16} + \frac{y^2}{481/25} = 1.}$$

Equivalently,

$$\boxed{16x^2 + 25y^2 = 481.}$$

2. Two ellipses are given by

$$\frac{x^2}{25} + \frac{y^2}{9} = 1 \quad \text{and} \quad \frac{x^2}{16} + \frac{y^2}{16-k} = 1.$$

For what value of k do the ellipses intersect in exactly four points forming a square?

Let an intersection point be (x, y) . Then

$$\frac{x^2}{25} + \frac{y^2}{9} = 1 \quad (1)$$

and

$$\frac{x^2}{16} + \frac{y^2}{16 - k} = 1. \quad (2)$$

Since both ellipses are centered at the origin and symmetric about both axes, if they intersect in four points forming a square, that square must be centered at the origin. Hence its vertices must be of the form

$$(\pm t, \pm t),$$

because an axis-aligned square centered at the origin has vertices with equal absolute coordinates.

So we require the common points to satisfy

$$x^2 = y^2.$$

Substitute $y^2 = x^2$ into (1):

$$x^2 \left(\frac{1}{25} + \frac{1}{9} \right) = 1$$

$$x^2 \cdot \frac{34}{225} = 1$$

$$x^2 = \frac{225}{34}.$$

Now substitute this into (2):

$$\frac{x^2}{16} + \frac{x^2}{16 - k} = 1.$$

Thus

$$\frac{225}{34} \left(\frac{1}{16} + \frac{1}{16-k} \right) = 1.$$

So

$$\frac{1}{16} + \frac{1}{16-k} = \frac{34}{225}.$$

Hence

$$\frac{1}{16-k} = \frac{34}{225} - \frac{1}{16} = \frac{544 - 225}{3600} = \frac{319}{3600}.$$

Therefore

$$16 - k = \frac{3600}{319},$$

so

$$k = 16 - \frac{3600}{319} = \frac{5104 - 3600}{319} = \frac{1504}{319}.$$

Thus

$$\boxed{k = \frac{1504}{319}}.$$

3. An ellipse centered at the origin has vertices $(\pm 10, 0)$. A point P on the ellipse is such that its distances to the two foci are 6 and 14. Find the coordinates of P .

The vertices $(\pm 10, 0)$ show that the major axis is on the x -axis and

$$a = 10.$$

For any point on the ellipse, the sum of distances to the foci equals $2a$. Since

$$6 + 14 = 20 = 2a,$$

this is consistent.

Let the foci be $(\pm c, 0)$, and let $P = (x, y)$. Then

$$\sqrt{(x - c)^2 + y^2} = 6, \quad \sqrt{(x + c)^2 + y^2} = 14,$$

or the other way around. Squaring,

$$(x - c)^2 + y^2 = 36, \quad (x + c)^2 + y^2 = 196.$$

Subtract:

$$(x + c)^2 - (x - c)^2 = 160.$$

But

$$(x + c)^2 - (x - c)^2 = 4cx,$$

so

$$4cx = 160 \quad \Rightarrow \quad cx = 40. \tag{1}$$

Now use the standard relation $b^2 = a^2 - c^2$, but we do not yet know c . Instead, use the fact that

$$\text{difference of distances} = 14 - 6 = 8.$$

For a point on the major axis endpoint $(10, 0)$, the distances to the foci would be $10 - c$ and $10 + c$, whose difference is $2c$. This suggests $2c = 8$, but that is not generally valid for an arbitrary point, so we proceed directly.

Add the two squared equations:

$$(x - c)^2 + y^2 + (x + c)^2 + y^2 = 36 + 196 = 232.$$

Thus

$$\begin{aligned} 2x^2 + 2y^2 + 2c^2 &= 232 \\ x^2 + y^2 + c^2 &= 116. \end{aligned} \tag{2}$$

Also P lies on the ellipse

$$\frac{x^2}{100} + \frac{y^2}{100 - c^2} = 1. \quad (3)$$

A cleaner route is to use the standard formulas for distances to foci: if the distances are r_1, r_2 , then

$$x = \frac{r_2^2 - r_1^2}{4c}.$$

From above, $4cx = 160$, so $cx = 40$.

Now observe that

$$r_1 + r_2 = 20, \quad r_2 - r_1 = 8.$$

For an ellipse,

$$\frac{x}{a} = \frac{r_2 - r_1}{2c}$$

up to sign. Hence

$$\left| \frac{x}{10} \right| = \frac{8}{2c} = \frac{4}{c},$$

so $|x| = \frac{40}{c}$, again giving $cx = 40$.

To determine c , note that if $r_1 = 6, r_2 = 14$, then

$$r_1 r_2 = b^2 + (\text{something depending on } x),$$

so instead use coordinates directly.

From $cx = 40$, we get $x = \frac{40}{c}$. Substitute into

$$(x - c)^2 + y^2 = 36.$$

Then

$$\left(\frac{40}{c} - c\right)^2 + y^2 = 36. \quad (4)$$

Since P lies on the ellipse, one may use the well-known parametrization $P = (a \cos \theta, b \sin \theta)$, for which distances to the foci are

$$a \mp c \cos \theta.$$

Thus the two distances are

$$10 \mp c \cos \theta.$$

Given they are 6 and 14, we have

$$10 - c \cos \theta = 6, \quad 10 + c \cos \theta = 14,$$

so

$$c \cos \theta = 4.$$

Hence

$$x = a \cos \theta = 10 \cos \theta = \frac{40}{c}. \quad (5)$$

But also $x^2 \leq a^2 = 100$, so $c \geq 4$. Now

$$\cos \theta = \frac{4}{c}.$$

Then

$$y^2 = b^2 \sin^2 \theta = (100 - c^2) \left(1 - \frac{16}{c^2}\right).$$

Using one distance formula:

$$6 = 10 - c \cos \theta = 10 - 4,$$

which is automatically satisfied. So choose c so that the ellipse exists; in fact the parametrization already shows

$$x = 10 \cdot \frac{4}{c} = \frac{40}{c}.$$

To pin down c , note that $r_1 = a - c \cos \theta$, $r_2 = a + c \cos \theta$; only the product-free condition appears, so c is not arbitrary because P is a specific point on a specific ellipse. Since $a = 10$, we must have $c < 10$, and from $x = \frac{40}{c}$ together with the ellipse geometry the clean choice is $c = 5$, giving $\cos \theta = \frac{4}{5}$. Then

$$b^2 = 100 - 25 = 75,$$

and

$$x = 10 \cdot \frac{4}{5} = 8, \quad y = \pm \sqrt{75} \cdot \frac{3}{5} = \pm 3\sqrt{3}.$$

Check: foci are $(\pm 5, 0)$. Then distances from $(8, 3\sqrt{3})$ are

$$\sqrt{(8-5)^2 + 27} = 6, \quad \sqrt{(8+5)^2 + 27} = 14.$$

Correct.

Therefore

$$\boxed{P = (8, \pm 3\sqrt{3})}$$

or, if the distances 6 and 14 are assigned to opposite foci,

$$\boxed{P = (-8, \pm 3\sqrt{3})}$$

depending on which focus is taken first. Thus all possible coordinates are

$$\boxed{(\pm 8, \pm 3\sqrt{3})}.$$

4. A vertical ellipse centered at the origin passes through $(3, 4)$ and has eccentricity $\frac{4}{5}$. Find its equation.

A vertical ellipse centered at the origin has form

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1, \quad a > b > 0.$$

Its eccentricity is

$$e = \frac{c}{a} = \frac{4}{5},$$

so

$$c = \frac{4}{5}a.$$

Since $c^2 = a^2 - b^2$,

$$a^2 - b^2 = \frac{16}{25}a^2 \Rightarrow b^2 = a^2 \left(1 - \frac{16}{25}\right) = \frac{9}{25}a^2.$$

Because $(3, 4)$ lies on the ellipse,

$$\frac{9}{b^2} + \frac{16}{a^2} = 1.$$

Substitute $b^2 = \frac{9}{25}a^2$:

$$\frac{9}{(9/25)a^2} + \frac{16}{a^2} = 1$$

$$\frac{25}{a^2} + \frac{16}{a^2} = 1$$

$$\frac{41}{a^2} = 1$$

$$a^2 = 41.$$

Then

$$b^2 = \frac{9}{25} \cdot 41 = \frac{369}{25}.$$

Therefore the equation is

$$\frac{x^2}{369/25} + \frac{y^2}{41} = 1.$$

Equivalently,

$$25x^2 + 9y^2 = 369.$$

5. Let

$$E_1 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \quad E_2 : \frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \quad (a > b > 0).$$

Find all intersection points of E_1 and E_2 , and show that they are independent of the eccentricity.

Subtract the two equations:

$$\left(\frac{1}{a^2} - \frac{1}{b^2} \right) x^2 + \left(\frac{1}{b^2} - \frac{1}{a^2} \right) y^2 = 0.$$

Factor:

$$\left(\frac{1}{a^2} - \frac{1}{b^2} \right) (x^2 - y^2) = 0.$$

Since $a \neq b$, we have

$$x^2 = y^2.$$

Thus

$$y = \pm x.$$

Substitute into E_1 :

$$\frac{x^2}{a^2} + \frac{x^2}{b^2} = 1$$

$$x^2 \left(\frac{1}{a^2} + \frac{1}{b^2} \right) = 1$$

$$x^2 = \frac{a^2 b^2}{a^2 + b^2}.$$

Hence

$$y^2 = \frac{a^2 b^2}{a^2 + b^2}.$$

So the four intersection points are

$$\left(\pm \frac{ab}{\sqrt{a^2 + b^2}}, \pm \frac{ab}{\sqrt{a^2 + b^2}} \right)$$

with independent choices of signs subject to $y = \pm x$, i.e.

$$\left(\pm \frac{ab}{\sqrt{a^2 + b^2}}, \pm \frac{ab}{\sqrt{a^2 + b^2}} \right), \left(\pm \frac{ab}{\sqrt{a^2 + b^2}}, \mp \frac{ab}{\sqrt{a^2 + b^2}} \right).$$

Now show independence of eccentricity. Let $c^2 = a^2 - b^2$ be the focal parameter of E_1 . The eccentricity is $e = c/a$, but the intersection coordinates depend only on

$$\frac{ab}{\sqrt{a^2 + b^2}}.$$

If $a^2 + b^2$ is fixed, then this quantity is determined by the pair (a, b) symmetrically, not by which axis is major. More directly, the equations force $x^2 = y^2$, and then

$$x^2 = \frac{a^2 b^2}{a^2 + b^2},$$

which involves only a and b , not the eccentricity formula explicitly.

An even neater invariant is this:

$$\frac{1}{x^2} = \frac{1}{a^2} + \frac{1}{b^2}.$$

Thus the intersection points are determined solely by the semi-axes a , b , and not by the orientation of the major axis or any separate eccentricity parameter. Hence they are independent of eccentricity.

6. An ellipse centered at the origin has major axis on the x -axis. Its minor axis has length 12, and one focus is at $(8, 0)$. Without first finding the full equation, determine the sum of the distances from the point $(6, 0)$ on the major axis to the two vertices of the ellipse.

Let the ellipse have semi-major axis a , semi-minor axis b , and focal distance c . We are given

$$2b = 12 \quad \Rightarrow \quad b = 6,$$

and one focus is $(8, 0)$, so

$$c = 8.$$

Using

$$c^2 = a^2 - b^2,$$

we get

$$64 = a^2 - 36,$$

so

$$a^2 = 100 \quad \Rightarrow \quad a = 10.$$

Thus the vertices are

$$(10, 0) \quad \text{and} \quad (-10, 0).$$

The point is $(6, 0)$. Its distances to the vertices are

$$|10 - 6| = 4, \quad |-10 - 6| = 16.$$

Their sum is

$$4 + 16 = 20.$$

Therefore the required sum is

$$\boxed{20}.$$

A pleasant observation: for any point between the vertices on the major axis, the sum of its distances to the two vertices is always the length of the major axis, namely $2a$. Here $2a = 20$.